

3. The following table contains data about four of the planets that orbit the Sun.

	Distance from Sun (million km)	Surface temperature (°C)	Diameter (units)	Mass (units)	Gravitational field strength, g (N/kg)
Earth	150	15	10	60	10.0
Mars	230	−65	5	6	3.7
Saturn	1 435	−140	100	5 700	9.0
Venus	108	464	10	49	9.0

(adapted from planetary fact sheet NASA)

Use only information from the table to answer the following questions.

(a) Underline the correct word in each of the sentences below. [3]

(i) The planet nearest the Sun is (**Earth** / Mars / Saturn / Venus).

(ii) The coldest planet is (**Earth** / Mars / Saturn / Venus).

(iii) The biggest planet is (**Earth** / Mars / Saturn / Venus).

(b) (i) State the gravitational field strength on Venus. [1]

..... N/kg

(ii) Use the equation:

$$\text{weight} = \text{mass} \times \text{gravitational field strength on Venus}$$

to calculate the weight of an object of mass 80 kg on Venus. [2]

weight = N

(iii) Name a planet on which an object of mass 80 kg would have the same weight as on Venus. [1]

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- (c) One student, Ceri, says that planets with the same diameter have the same mass.
Explain whether you agree with this statement.

[2]

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